United States Patent Application Publication	20250271561
Kind Code	A1
Publication Date	August 28, 2025
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METHOD FOR ATTENUATING EFFECTS OF GALVANOMETER ANGLE MEASUREMENT NOISE IN SCANNER POINT CLOUD

Abstract

A beam scanning method includes generating a first driving signal based on a first angle setpoint; generating a second driving signal based on a second angle setpoint; driving a two-dimensional scanner about a first scanning axis based on the first driving signal and about a second scanning axis based on the second driving signal; generating a distance measurement based on a reflected light beam; generating a first estimated angle measurement signal based on the first angle setpoint and a dynamic model of the two-dimensional scanner; generating a second estimated angle measurement signal based on the second angle setpoint and the dynamic model of the two-dimensional scanner; associating the distance measurement with a first estimated angle value corresponding to the first estimated angle measurement signal; and associating the distance measurement with a second estimated angle value corresponding to the second estimated angle measurement signal.

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Family ID: **1000007954207**

Appl. No.: **18/749822**

Filed: **June 21, 2024**

Related U.S. Application Data

us-provisional-application US 63558960 20240228

Publication Classification

Int. Cl.: **G01S7/497** (20060101); **G01S17/894** (20200101)

U.S. Cl.:

CPC **G01S7/497** (20130101); **G01S17/894** (20200101);

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION [0001] This patent application claims priority to U.S. Patent Application No. 63/558,960, filed on Feb. 28, 2024, and entitled "METHOD FOR ATTENUATING EFFECTS OF GALVANOMETER ANGLE MEASUREMENT NOISE IN SCANNER POINT CLOUD." The disclosure of the prior application is considered part of and is incorporated by reference into this patent application.

TECHNICAL FIELD
[0002] The present disclosure relates generally to beam scanning systems and methods for beam scanning.
BACKGROUND
[0003] A scanning system may use two-dimensional (2D) or three-dimensional (3D) scanning to scan one or more light beams within a field-of-view (FOV) according to a scanning pattern. The scanning system may use two scanning axes, including a first scanning axis that is configured to steer the one or more light beams in a first direction at a first scanning frequency and a second scanning axis that is configured to steer the one or more light beams in a second direction at a second scanning frequency. The second scanning axis is typically perpendicular to the first scanning axis. Transmitted light beams may be reflected back to the scanning system from one or more objects in the FOV as reflected light beams. A 3D image of a scanned scene or a scanned object can then be generated based on distance measurements corresponding to the transmitted/reflected light beams. Additionally, or alternatively, the reflected light beams may be used by the scanning system to detect objects within the FOV for further processing.
SUMMARY
[0004] In some implementations, a beam scanning system includes a two-dimensional scanner comprising a first galvanometer scanner configured to rotate about a first scanning axis based on a first driving signal, and a second galvanometer scanner configured to rotate about a second scanning axis based on a second driving signal; a time-of-flight sensor configured to receive a reflected light beam and generate a distance measurement based on the reflected light beam; a driver system configured to receive a first angle setpoint for the first galvanometer scanner and a second angle setpoint for the second galvanometer scanner, drive the first galvanometer scanner with the first driving signal based on the first angle setpoint, and drive the second galvanometer scanner with the second driving signal based on the second angle setpoint; and a system controller configured with a dynamic model of the two-dimensional scanner, wherein the system controller is configured to generate a first estimated angle measurement signal based on the first angle setpoint and the dynamic model, wherein the first estimated angle measurement signal follows a first angular trajectory of the first galvanometer scanner about the first scanning axis, wherein the system controller is configured to generate a second estimated angle measurement signal based on the second angle setpoint and the dynamic model, wherein the second estimated angle measurement signal follows a second angular trajectory of the second galvanometer scanner about the second scanning axis, and wherein the system controller is configured to associate the distance measurement with a first estimated angle value corresponding to the first estimated angle measurement signal, associate the distance measurement with a second estimated angle value corresponding to the second estimated angle measurement signal, and generate a point in a three-dimensional point cloud based on the distance measurement, the first estimated angle value, and the

second estimated angle value.

[0005] In some implementations, a beam scanning system includes a two-dimensional scanner comprising a galvanometer scanner configured to rotate about a first scanning axis based on a first driving signal, and rotate about a second scanning axis based on a second driving signal; a time-of-flight sensor configured to receive a reflected light beam and generate a distance measurement based on the reflected light beam; a driver system configured to receive an angle vector setpoint corresponding to a two-dimensional scanning coordinate, drive the galvanometer scanner about the first scanning axis with the first driving signal based on the angle vector setpoint, and drive the galvanometer scanner about the second scanning axis with the second driving signal based on the angle vector setpoint; and a system controller configured with a dynamic model of the two-dimensional scanner, wherein the system controller is configured to generate a first estimated angle measurement signal based on the angle vector setpoint and the dynamic model, wherein the first estimated angle measurement signal follows a first angular trajectory of the galvanometer scanner about the first scanning axis, wherein the system controller is configured to generate a second estimated angle measurement signal based on the angle vector setpoint and the dynamic model, wherein the second estimated angle measurement signal follows a second angular trajectory of the galvanometer scanner about the second scanning axis, and wherein the system controller is configured to associate the distance measurement with a first estimated angle value corresponding to the first estimated angle measurement signal, associate the distance measurement with a second estimated angle value corresponding to the second estimated angle measurement signal, and generate a point in a three-dimensional point cloud based on the distance measurement, the first estimated angle value, and the second estimated angle value.

[0006] In some implementations, a beam scanning method includes generating a first driving signal based on a first angle setpoint; generating a second driving signal based on a second angle setpoint; driving a two-dimensional scanner about a first scanning axis based on the first driving signal and about a second scanning axis based on the second driving signal; generating a distance measurement based on a reflected light beam; generating a first estimated angle measurement signal based on the first angle setpoint and a dynamic model of the two-dimensional scanner, wherein the first estimated angle measurement signal follows a first angular trajectory about the first scanning axis; generating a second estimated angle measurement signal based on the second angle setpoint and the dynamic model of the two-dimensional scanner, wherein the second estimated angle measurement signal follows a second angular trajectory about the second scanning axis; associating the distance measurement with a first estimated angle value corresponding to the first estimated angle measurement signal; associating the distance measurement with a second estimated angle value corresponding to the second estimated angle measurement signal; and generating a point in a three-dimensional point cloud based on the distance measurement, the first estimated angle value, and the second estimated angle value.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0007] FIG. **1**A is a schematic block diagram of a 2D scanning system according to one or more implementations.

[0008] FIG. **1**B is a schematic block diagram of a 2D scanning system according to one or more implementations.

[0009] FIG. **2** shows a control loop of a beam scanning system according to one or more implementations.

[0010] FIG. **3** shows a modeling system according to one or more implementations.

[0011] FIG. **4** shows a system according to one or more implementations.

[0012] FIG. **5** shows a system according to one or more implementations.

[0013] FIG. **6** is a flowchart of an example process associated with a beam scanning method.

DETAILED DESCRIPTION

[0014] The following detailed description of example implementations refers to the accompanying drawings. The same reference numbers in different drawings may identify the same or similar elements.

[0015] In 3D sensing and imaging technologies, such as light detection and ranging (LIDAR), a scan can be performed to illuminate an area referred to as a field-of-view. For example, a scanning mirror may be arranged to receive transmitted light beams from a light transmitter and steer (scan) the transmitted light beams into the field-of-view to perform a scanning of the environment. The transmitted light beams may be backscattered by one or more objects back toward the scanning system as reflected light beams where the reflected light beams are detected by a sensor. For example, the sensor may be a photodetector array. The sensor may convert each reflected light beam into an electrical signal (e.g., a current signal or a voltage signal) that may be further processed by the scanning system to generate object data or an image, such as a point cloud. While light may be scanned in two dimensions, a third dimension (e.g., a depth dimension) may be obtained from distance measurements. The distance measurements may be performed based on a time-of-flight of transmitted and reflected light beams. 2D scanning coordinates along with distance information may be used to generate a point cloud of a scanned environment.

[0016] A single scanning mirror with two scanning axes may be used for a scanning operation to perform a two-dimensional scan. Alternatively, two scanning mirrors, each with a single scanning axis, may be used for a scanning operation to perform the two-dimensional scan. For example, the scanning axes of the two scanning mirrors may be arranged orthogonal to each other such that one scanning mirror scans in a x-direction and the other scanning mirror scans in a y-direction. Thus, the two scanning mirrors are used in combination to direct a laser beam in a two-dimensional scanning pattern, such as a raster pattern, to scan the field-of-view. The laser beam may be reflected off an object and the returned reflected beam is analyzed to determine the time-of-flight which provides a third measurement dimension (e.g., a distance measurement to the object). The system may measure the mirror angle of each scanning mirror, and the measured mirror angles of the two scanning mirrors may be combined with the distance measurement. These three measurements may be processed into a final 3D point cloud. However, if the mirror angle measurements are corrupted by noise, the final point cloud will be degraded and less accurate, which may negatively affect object detection and/or may require more complex signal processing that requires higher processing bandwidth and/or power. In some cases, the noise may render the final point cloud unusable.

[0017] Some implementations provide a beam scanning system with a galvanometer-based two-dimensional scanner and a method for attenuating an angle measurement noise without attenuating an angle signal. The galvanometer-based two-dimensional scanner may include a single galvanometer scanner (e.g., a single scanning mirror) with two scanning axes, or may include two galvanometer scanners (e.g., two scanning mirrors), each with a single scanning axis. A galvanometer scanner may have a rapidly moving angle has a high frequency spectral content. As a result, an angle measurement cannot simply be low-pass filtered to remove broadband measurement noise, as this would also remove important information from an angle measurement signal, resulting in point cloud inaccuracy. Instead, one or more implementations may use a dynamic model of the galvanometer-based two-dimensional scanner to remove the angle measurement noise in order to provide more accurate angle measurements. As a result, the beam scanning system may generate a more accurate 3D point cloud based on the angle measurements.

[0018] FIG. **1**A is a schematic block diagram of a 2D scanning system **100**A according to one or more implementations. In particular, the 2D scanning system **100**A includes a scanner **102** configured to steer or otherwise deflect light beams according to a 2D scanning pattern for scanning 3D objects. The 2D scanning system **100**A further includes a driver system **104**, a system controller **106**, and a light transmitter **108**, and a sensor **110**.

[0019] In the example shown in FIG. **1**A, the scanner **102** may be a mechanical moving mirror and may be configured to rotate or oscillate via rotation about two scanning axes that are typically orthogonal to each other. For example, the two scanning axes may include a first scanning axis **112** that enables the scanner **102** to steer light in a first scanning direction (e.g., an x-direction) and a second scanning axis **114** (e.g., an inner scanning axis) that enables the scanner **102** to steer light in a second scanning direction (e.g., a y-direction). As a result, the scanner **102** can direct light beams in two dimensions according to the 2D scanning pattern.

[0020] A scan can be performed to illuminate an area referred to as a field-of-view. The scan, such as an oscillating horizontal scan (e.g., from left to right and right to left of a field-of-view), an oscillating vertical scan (e.g., from bottom to top and top to bottom of a field-of-view), or a combination thereof (e.g., a Lissajous scan or a raster scan) can illuminate the field-of-view in a continuous scan fashion. In some implementations, the 2D scanning system **100**A may be configured to transmit successive light beams (e.g., as successive light pulses) in different scanning directions to scan the field-of-view. The scanner **102** can direct a transmitted light beam at a desired 2D measurement coordinate (e.g., an x-y coordinate) in the field-of-view, controlled by the system controller **106**.

[0021] In some implementations, the scanner **102** may be arranged to receive transmitted light beams from the light transmitter **108** and steer (scan) the transmitted light beams into the field-of-view to perform a scanning of the environment. The transmitted light beams may be backscattered by one or more objects back toward the 2D scanning system **100**A as reflected light beams, where the reflected light beams are detected by the sensor **110**. For example, the sensor **110** may be a photodetector array. The sensor **110** may convert each reflected light beam into an electrical signal (e.g., a current signal or a voltage signal) that may be further processed by the 2D scanning system **100**A to generate object data or an image. Thus, the sensor **110** may be a time-of-flight sensor configured to receive a reflected light beam and generate one or more distance measurements based on the reflected light beam. In such implementations, the desired 2D measurement coordinate may correspond to a particular transmission direction in the field-of-view that is targeted by the transmitted light beam for object detection or scanning, with different 2D measurement coordinates corresponding to different transmission directions. The system controller **106** may receive electrical signals from the sensor and perform signal processing on the electrical signals for object feature detection.

[0022] Accordingly, multiple light beams transmitted at different transmission times can be steered by the scanner **102** at the different 2D measurement coordinates of the field-of-view in accordance with the 2D scanning pattern. The scanner **102** can be used to scan the field-of-view in both scanning directions by changing an angle of deflection of the scanner **102** on each of the first scanning axis **112** and the second scanning axis **114**.

[0023] The driver system **104** may be configured to generate driving signals (e.g., actuation signals) to drive the scanner **102** about the first scanning axis **112** and the second scanning axis **114**. In particular, the driver system **104** may be configured to apply the driving signals to an actuator structure of the scanner **102**. In some implementations, the driver system **104** includes a driver **116** configured to drive the scanner **102** about the first scanning axis **112** and the second scanning axis **114**. The scanner **102** may have separate actuator structures for each scanning axis. Thus, the scanner **102** may have a first actuator structure for the first scanning axis **112**, and a second actuator structure for the second scanning axis **114**. The driver **116** may apply a first driving signal to the first actuator structure to drive the scanner **102** about the first scanning axis, and may apply a second driving signal to the second actuator structure to drive the scanner **102** about the second scanning axis. In some implementations, the driver **116** may include separate drivers for each scanning axis **112** and **114**. In implementations in which the scanner **102** is used as an oscillator, the driver **116** may be configured to drive an oscillation of the scanner **102** about the first scanning axis **112** at a first frequency, and drive an oscillation of the scanner **102** about the second scanning axis **114** at a second frequency.

[0024] The driver **116** may be configured to receive feedback information from the scanner **102**, such as rotational position information (e.g., an angle measurement). The system controller **106** may use the rotational position information to trigger light beams at the light transmitter **108**. For example, the system controller **106** may use the rotational position information to set a transmission time of light transmitter **108** in order to target a particular 2D measurement coordinate of the 2D scanning pattern.

[0025] In some implementations, the system controller **106** is configured to set a driving frequency of the scanner **102** for each scanning axis and is capable of synchronizing the oscillations about the first scanning axis **112** and the second scanning axis **114**. In particular, the system controller **106** may be configured to control an actuation of the scanner **102** about each scanning axis by controlling the driving signals. The system controller **106** may control the frequency, the phase, the duty cycle, and/or a voltage level of the driving signals to control the actuations about the first scanning axis **112** and the second scanning axis **114**. The actuation of the scanner **102** about a particular scanning axis controls its range of motion and scanning rate about that particular scanning axis.

[0026] The light transmitter **108** may include one or more light sources, such as one or more laser diodes or one or more light emitting diodes, for generating one or more light beams. In some implementations, the light transmitter **108** may be configured to sequentially transmit a plurality of light beams (e.g., light pulses) as the scanner **102** changes its transmission direction in order to target different 2D measurement coordinates. The plurality of light beams may include visible light, infrared (IR) light, or other types of illumination signals, depending on an application of the 2D scanning system **100**A. A transmission sequence of the plurality of light beams and a timing thereof may be implemented by the light transmitter **108** according to a control signal CTRL received from the system controller **106**.

[0027] The system controller **106** may be configured to control components of the 2D scanning system **100**A. In certain applications, the system controller **106** may also be configured to receive programming information with respect to the 2D scanning pattern and control a timing of the plurality of light beams generated by the light transmitter **108** based on the programming information. Thus, the system controller **106** may include both processing and control circuitry that is configured to generate control signals for controlling the light transmitter **108** and the driver **116**. For example, the system controller **106** may include processing circuitry **118** configured to execute machine instructions, and, based on executing the machine instructions, generate control signals for controlling the 2D scanning system **100**A to perform a 2D scan of the scanning area according to the 2D scanning pattern. Thus, the processing circuitry may include one or more processors and other signal processing components. In some implementations, the processing circuitry **118** may include a DSP. The processing circuitry **118**, in conjunction with control circuitry, may control the light transmitter **108** and the scanner **102** to target each 2D measurement coordinate with a respective light beam. The processing circuitry **118** may control the scanner **102** by controlling one or more parameters of the driver **116**, such as the frequency, the phase, the duty cycle, and/or a voltage level of the driving signals used for driving each scanning axis **112** and **114**. The processing circuitry **118** may process a plurality of measurements signals and generate a 3D point cloud based on the plurality of measurement signals. In some implementations, in which the plurality of light beams is used, the system controller **106** may be configured to generate the control signal CTRL used for triggering the light transmitter **108** to generate the plurality of light beams. Using the control signal CTRL, the system controller **106** can control the transmission times of the plurality of light beams of the light transmitter **108** to achieve a desired illumination pattern within the field-of-view. The desired illumination pattern is produced by a combination of the 2D scanning pattern produced by the scanner **102** and the transmission times triggered by the system controller **106**.

[0028] Accordingly, the 2D scanning system **100**A may include a detector that includes at least one sensor (e.g., sensor **110**) and at least one signal processor (e.g., the processing circuitry **118** or other additional processors and/or processing components) implemented, for example, in the system controller **106**. The sensor **110** may generate electrical signals based on reflected light beams corresponding to the light beams transmitted by the light transmitter **108**. The sensor **110** may transmit the electrical signals to processing circuitry **118**. The processing circuitry **118** may be configured to process the electrical signals to generate distance measurements based on the machine instructions for generating the 3D point cloud.
[0029] In some implementations, the scanner **102** may be a galvanometer scanner. The galvanometer scanner may include a shaft for each scanning axis, a first galvanometer-based scanning motor that drives a rotation of a first shaft associated with the first scanning axis **112**, a second galvanometer-based scanning motor that drives a rotation of a second shaft associated with the second scanning axis **114**, an optical mirror mounted to both the first shaft and the second shaft, and a detector that provides positional feedback (e.g., an actual angle measurement for each scanning axis or a vector measurement) to the system controller **106**. The driver system **104** may include a first servo driver for driving the first galvanometer-based scanning motor, and a second servo driver for driving the second galvanometer-based scanning motor. Each servo driver may generate a driving signal (e.g., a drive current) based on a command position (e.g., an angle setpoint) that is provided to the servo driver by a control loop. Each servo driver may supply the driving signal to a respective galvanometer-based scanning motor. The system controller **106** may monitor a difference representing an error between the command position (e.g., the angle setpoint) and an actual position (e.g., the actual angle measurement) to adjust the command position based on the difference.
[0030] The driver system **104** may receive, from the system controller **106**, an angle vector setpoint corresponding to a two-dimensional scanning coordinate, drive the scanner **102** about the first scanning axis **112** with the first driving signal based on the angle vector setpoint, and drive the scanner **102** about the second scanning axis **114** with the second driving signal based on the angle vector setpoint. The angle vector setpoint may include a first angle setpoint for the first scanning axis **112** and a second angle setpoint for the second scanning axis **114**.
[0031] The system controller **106** may be configured with a dynamic model of the scanner **102**. The system controller **106** may generate a first estimated angle measurement signal based on the angle vector setpoint and the dynamic model. The first estimated angle measurement signal may a first angular trajectory of the scanner **102** about the first scanning axis **112**. Additionally, the system controller **106** may generate a second estimated angle measurement signal based on the angle vector setpoint and the dynamic model. The second estimated angle measurement signal may follow a second angular trajectory of the scanner **102** about the second scanning axis **114**. Since the first estimated angle measurement signal and the second estimated angle measurement signal are generated based on the dynamic model, the first estimated angle measurement signal and the second estimated angle measurement signal may be free of or substantially free of angle measurement noise. In other words, the system controller **106** may remove angle measurement noise based on the dynamic model to generate the first estimated angle measurement signal and the second estimated angle measurement signal.
[0032] In some implementations, the system controller **106** may generate estimated angle vector measurement signal based on the angle vector setpoint, an angle measurement vector signal, and the dynamic model. The estimated angle vector measurement signal may represent a combination of the first estimated angle measurement signal and the second estimated angle measurement signal.
[0033] In some implementations, the driver system **104** may include a first angle position detector configured to generate a first angle measurement signal based on detecting a first angular position of the scanner **102** about the first scanning axis **112**, and a second angle position detector

configured to generate a second angle measurement signal based on detecting a second angular position of the scanner **102** about the second scanning axis **114**. The angle measurement vector signal may represent a combination of the first angle measurement signal and the second angle measurement signal. The driver system **104** may provide the first angle measurement signal and the second angle measurement signal to the system controller **106** as positional feedback.

[0034] The system controller **106** may also receive distance measurements from the sensor **110**. The system controller **106** may associate a distance measurement with a first estimated angle value corresponding to the first estimated angle measurement signal, associate the distance measurement with a second estimated angle value corresponding to the second estimated angle measurement signal, and generate a point in a 3D point cloud based on the distance measurement, the first estimated angle value, and the second estimated angle value. Since the first estimated angle measurement signal and the second estimated angle measurement signal are free of or substantially free of angle measurement noise, a 3D-coordinate of the point within the 3D point cloud may be more accurate than would otherwise be possible if the angle measurement noise were still present.

[0035] As indicated above, FIG. **1** is provided as an example. Other examples may differ from what is described with regard to FIG. **1**. In practice, the 2D scanning system **100**A may include additional components, fewer components, different components, or differently arranged components than those shown in FIG. **1** without deviating from the disclosure provided above. In addition, in some implementations, the 2D scanning system **100**A may include one or more additional mirrors to scan the field-of-view.

[0036] FIG. **1**B is a schematic block diagram of a 2D scanning system **100**B according to one or more implementations. In particular, the 2D scanning system **100**B includes two scanners, a first scanner **102***x* and a second scanner **102***y*, that are optically coupled in series to steer or otherwise deflect light beams according to a 2D scanning pattern. The first scanner **102***x* and the second scanner **102***y* are similar to the scanner **102** described in FIG. **1**A, with the exception that the first scanner **102***x* and the second scanner **102***y* are configured to rotate about a single scanning axis instead of two scanning axes. The first scanner **102***x* is configured to rotate about the first scanning axis **112** to steer light in the x-direction, and the second scanner **102***y* is configured to rotate about the second scanning axis **114** to steer light in the y-direction. Similar to the scanner **102** described in FIG. **1**A, the first scanner **102***x* and the second scanner **102***y* may be galvanometer scanners configured to be driven by a respective galvanometer-based scanning motor.

[0037] Because each of the first scanner **102***x* and the second scanner **102***y* is configured to rotate about a single scanning axis, each of the first scanner **102***x* and the second scanner **102***y* is responsible for scanning light in one dimension. As a result, the first scanner **102***x* and the second scanner **102***y* may be referred to as one-dimensional (1D) scanner. In the example shown in FIG. **1**B, the first scanner **102***x* and the second scanner **102***y* are used together to steer light beams in two dimensions. The first scanner **102***x* and the second scanner **102***y* are arranged sequentially along a transmission path of the light beams such that one of the scanners (e.g., the first scanner **102***x*) first receives a light beam and steers the light beam in a first dimension, and the second one of the scanners (e.g., the second scanner **102***y*) receives the light beam from the first scanner **102***x* and steers the light beam in a second dimension. As a result, the first scanner **102***x* and the second scanner **102***y* operate together to steer the light beam generated by the light transmitter **108** in two dimensions. In this way, the first scanner **102***x* and the second scanner **102***y* can direct the light beam at a desired 2D coordinate (e.g., an x-y coordinate) in the field-of-view. Multiple light beams can be steered by the first scanner **102***x* and the second scanner **102***y* at different 2D coordinates of a 2D scanning pattern.

[0038] The driver system **104**, the system controller **106**, and the light transmitter **108** are configured to operate as similarly described above in reference to FIG. **1**A. The driver **116** may be electrically coupled to the first scanner **102***x* to drive the first scanner **102***x* about the first scanning axis **112**, and the driver system **104** may detect a position (e.g., an angular position) of the first scanner **102***x* about the first scanning axis **112** to provide first position information to the system controller **106**. Similarly, the driver **116** may be electrically coupled to the second scanner **102***y* to drive the second scanner **102***y* about the second scanning axis **114**, and the driver system **104** may detect a position (e.g., an angular position) of the second scanner **102***y* about the second scanning axis **114** to send a position of the second scanner **102***y* about the second scanning axis **114** to provide second position information to the system controller **106**.

[0039] In some implementations, the first scanner **102***x* may be a first galvanometer scanner configured to rotate about the first scanning axis **112** based on a first driving signal, and the second scanner **102***y* may be a second galvanometer scanner configured to rotate about the second scanning axis **114** based on a second driving signal.

[0040] A shaft may be provided for each scanning axis. The driver system **104** may include a first galvanometer-based scanning motor that drives a rotation of a first shaft associated with the first scanning axis **112**, a second galvanometer-based scanning motor that drives a rotation of a second shaft associated with the second scanning axis **114**, and detectors that provide positional feedback (e.g., an actual angle measurement for each scanning axis) to the system controller **106**. The driver system **104** may include a first servo driver for driving the first galvanometer-based scanning motor, and a second servo driver for driving the second galvanometer-based scanning motor. Each servo driver may generate a driving signal (e.g., a drive current) based on a command position (e.g., an angle setpoint) that is provided to the servo driver by a control loop. Each servo driver may supply the driving signal to a respective galvanometer-based scanning motor. The system controller **106** may monitor a difference representing an error between the command position (e.g., the angle setpoint) and an actual position (e.g., the actual angle measurement) to adjust the command position based on the difference.

[0041] The driver system **104** may receive, from the system controller **106**, a first angle setpoint for the first galvanometer scanner and a second angle setpoint for the second galvanometer scanner, drive the first galvanometer scanner with the first driving signal based on the first angle setpoint, and drive the second galvanometer scanner with the second driving signal based on the second angle setpoint.

[0042] The system controller **106** may be configured with a dynamic model of the two-dimensional scanner. The dynamic model of the two-dimensional scanner may include a first dynamic model of the first scanner **102***x*, and a second dynamic model of the second scanner **102***y*, or may be a combined dynamic model for both the first scanner **102***x* and the second scanner **102***y*. The system controller **106** may generate a first estimated angle measurement signal based on the first angle setpoint and the dynamic model (e.g., the first dynamic model or the combined dynamic model). The first estimated angle measurement signal may follow a first angular trajectory of the first scanner **102***x* about the first scanning axis **112**. Additionally, the system controller **106** may generate a second estimated angle measurement signal based on the second angle setpoint and the dynamic model (e.g., the second dynamic model or the combined dynamic model). The second estimated angle measurement signal may follow a second angular trajectory of the second scanner **102***y* about the second scanning axis **114**. Since the first estimated angle measurement signal and the second estimated angle measurement signal are generated based on the dynamic model of the two-dimensional scanner, the first estimated angle measurement signal and the second estimated angle measurement signal may be free of or substantially free of angle measurement noise. In other words, the system controller **106** may remove angle measurement noise based on the dynamic model to generate the first estimated angle measurement signal and the second estimated angle measurement signal.

[0043] In some implementations, the driver system **104** may include a first angle position detector configured to generate a first angle measurement signal based on detecting a first angular position of the first scanner **102***x* about the first scanning axis **112**, and a second angle position detector configured to generate a second angle measurement signal based on detecting a second angular position of the second scanner **102***y* about the second scanning axis **114**. The driver system **104** may provide the first angle measurement signal and the second angle measurement signal to the system controller **106** as positional feedback.

[0044] The system controller **106** may also receive distance measurements from the sensor **110**. The system controller **106** may associate the distance measurement with a first estimated angle value corresponding to the first estimated angle measurement signal, associate the distance measurement with a second estimated angle value corresponding to the second estimated angle measurement signal, and generate a point in a three-dimensional point cloud based on the distance measurement, the first estimated angle value, and the second estimated angle value. Since the first estimated angle measurement signal and the second estimated angle measurement signal are free of or substantially free of angle measurement noise, a 3D-coordinate of the point within the 3D point cloud may be more accurate than would otherwise be possible if the angle measurement noise were still present.

[0045] As indicated above, FIG. **1**B is provided as an example. Other examples may differ from what is described with regard to FIG. **1**B. In practice, the 2D scanning system **100**B may include additional components, fewer components, different components, or differently arranged components than those shown in FIG. **1**B without deviating from the disclosure provided above. In addition, in some implementations, the 2D scanning system **100**B may include one or more additional 1D MEMS mirrors or one or more additional light transmitters used to scan one or more additional field-of-views. Additionally, two or more components shown in FIG. **1**B may be implemented within a single component, or a single component shown in FIG. **1**B may be implemented as multiple, distributed components. Additionally, or alternatively, a set of components (e.g., one or more components) of the 2D scanning system **100**B may perform one or more functions described as being performed by another set of components of the 2D scanning system **100**B.

[0046] FIG. **2** shows a control loop **200** of a beam scanning system according to one or more implementations. The control loop **200** may be implemented for the scanner **102** described in connection with FIG. **1**A, the first scanner **102***x* described in connection with FIG. **1**B, or the second scanner **102***y* described in connection with FIG. **1**B. The control loop **200** may be duplicated for each scanning axis. In other words, separate control loops **200** may be provided for the first scanning axis **112** and the second scanning axis **114**.

[0047] The control loop **200** includes the driver **116** that generates a driving signal (e.g., a drive current), and a scanner **102***x,y* corresponding to one of the scanners described in connection with FIGS. **1**A and **1**B. The scanner **102***x,y* is driven about a scanning axis based on the driving signal. An angle position detector may be integrated with the scanner **102***x,y* and may be configured to detect an angular position (e.g., a rotation angle θ) of the scanner **102***x,y* about the scanning axis, and generate an actual angle measurement signal based on the angular position. The actual angle measurement signal provided by the angle position detector may represent an actual mirror angle θ of the scanner **102***x,y.*

[0048] A summer **202** may provide an angle measurement signal θmeas based on the actual angle measurement signal and angle measurement noise. In other words, the summer **202** may represent angle measurement noise being added to the actual angle measurement signal, which may corrupt the angle measurement signal (e.g., the actual mirror angle θ) to produce the angle measurement signal θmeas. Thus, the angle measurement signal θmeas may be a noisy measurement signal. For example, the summer **202** may represent any signal line, component, and/or combination of signal lines or components that introduce noise onto the actual angle measurement signal.

[0049] An error detector **204** may receive an angle setpoint θsp for the scanner **102***x,y*. The angle setpoint θsp may be a control value corresponding to a target angular position of the scanner **102***x,y*. The angle setpoint θsp may be provided in setpoint control signal that corresponds to a target angular trajectory of the scanner **102***x,y*. Additionally, the error detector **204** may receive the angle measurement signal θmeas, which may be corrupted by noise. The error detector **204** may generate an error signal that represents a difference between the angle setpoint θsp and the angle measurement signal θmeas. The driver **116** may adjust the driving signal in order to drive the error signal to zero. Thus, noise included in the angle measurement signal θmeas may cause the driver **116** to generate an incorrect driving signal.

[0050] Alternatively, the error detector **204** may be provided with an estimated angle measurement signal instead of the angle measurement signal θmeas provided by the summer **202**. The estimated angle measurement signal fest may be generated based on a dynamic model of the scanner **102***x,y*. As a result, the estimated angle measurement signal θest may be free of or substantially free of angle measurement noise, which may lead to more accurate control of the angular position of the scanner **102***x,y*. For example, the driver **116** may generate the driving signal based on a difference between the estimated angle measurement signal θest and the setpoint control signal. Thus, the driver **116** may compensate the driving signal based on a difference between the estimated angle measurement signal and the setpoint control signal. In other words, the driver may generate the driving signal based on the difference between the estimated angle measurement signal and the angle setpoint θsp (or angle vector setpoint θvsp) provided by the setpoint control signal.

[0051] In some implementations, the angle setpoint θsp may be an angle vector setpoint, the estimated angle measurement signal θest may be an estimated angle vector measurement signal that is based on the angle vector setpoint, and the angle measurement signal θmeas may be an angle measurement vector signal. The angle vector setpoint, the angle measurement vector signal, and/or the estimated angle vector measurement signal may be used to drive both scanning axes **112** and **114**. For example, angle vectors may be used in the 2D scanning system **100**A described in connection with FIG. **1**A. The first scanning axis **112** and the second scanning axis **114** of the scanner **102** (e.g., the 2D scanning mirror) may be cross-coupled by one or more cross-coupling effects or interactions. Thus, motion about the first scanning axis **112** may affect motion about the second scanning axis **114**, and vice versa. Thus, the angle vector setpoint, the angle measurement vector signal, and/or the estimated angle vector measurement signal may take into account the one or more cross-coupling effects or interactions between the scanning axes **112** and **114**.

[0052] As indicated above, FIG. **2** is provided as an example. Other examples may differ from what is described with regard to FIG. **2**.

[0053] FIG. **3** shows a modeling system **300** according to one or more implementations. The modeling system **300** may include a dynamic system identification processor **302** configured to generate a dynamic model of a two-dimensional scanner based on a plurality of input data. For example, the dynamic system identification processor **302** may receive a setpoint control signal and an angle measurement signal θmeas. The setpoint control signal may provide angle setpoints θsp corresponding to a target angular trajectory. The angle measurement signal θmeas may be provided by the summer **202** derived in connection with FIG. **2**. For each angle setpoints θsp, the dynamic system identification processor **302** may provide dynamic model parameter values that corresponds to a response or a behavior of the two-dimensional scanner. Thus, the dynamic system identification processor **302** may generate the dynamic model that models the behavior of the two-dimensional scanner for the entire target angular trajectory of the two-dimensional scanner. Since the angle measurement signal θmeas, which may be a noisy measurement signal, is used as an input, the dynamic model may take into account any noise that be present during dynamic system modeling, which may naturally occur during operation of the two-dimensional scanner.

[0054] The dynamic system identification processor **302** may execute a modeling algorithm, such as subspace identification, arx modeling, or back-box/grey-box model parametric optimization for performing the dynamic system modeling. In addition, model-fitting may include using normal scan measurement data as an input, or may use setpoints with specially tailored spectral content, for example, with filtered broadband noise, to improve a richness of the data used for the dynamic system modeling. The behavior of the two-dimensional scanner should not change significantly over time. Thus, the dynamic system modeling may be performed at periodic intervals or upon system startup. The modeling system **300** may be incorporated into the system controller **106** and may be performed by the processing circuitry **118**.

[0055] As indicated above, FIG. **3** is provided as an example. Other examples may differ from what is described with regard to FIG. **3**.

[0056] FIG. **4** shows a system **400** according to one or more implementations. The system **400** may include a first processing system **402** corresponding to the first scanning axis **112**, and a second processing system **404** corresponding to the second scanning axis **114**. Thus, the first processing system **402** and the second processing system **404** may be used in the 2D scanning system **100**A described in conjunction with FIG. **1**A or the 2D scanning system **100**B described in conjunction with FIG. **1**B.

[0057] The first processing system **402** may include a control loop **200***x* that is similar to the control loop **200** described in connection with FIG. **2**. The control loop **200***x* may be configured for driving a galvanometer scanner about the first scanning axis **112** and obtaining a first estimated angle measurement signal θest,x that may be provided to a processing component of the processing circuitry **118** for generating point cloud data. The galvanometer scanner may be the scanner **102** or the first scanner **102***x*. In the following examples, it will be assumed that the galvanometer scanner controlled by the control loop **200***x* corresponds to the first scanner **102***x*.

[0058] A driver system (e.g., driver system **104**) of the control loop **200***x* may receive a first angle setpoint θsp,x for the first scanner **102***x*, and drive the first scanner **102***x* with a first driving signal based on the first angle setpoint θsp,x. A first angle position detector of the control loop **200***x* may generate a first angle measurement signal θmeas,x based on detecting a first angular position of the first scanner **102***x* about the first scanning axis **112**. The first angle measurement signal θmeas,x may be a noisy measurement signal.

[0059] A processor **406**, implemented in the system controller **106**, may be configured with a dynamic model X that models the behavior of the first scanner **102***x* about the first scanning axis **112**. The processor **406** may generate the first estimated angle measurement signal θest,x based on the first angle setpoint θsp,x and the dynamic model X. The first estimated angle measurement signal θest,x may follow a first angular trajectory of the first scanner **102***x* about the first scanning axis **112**.

[0060] In some implementations, the processor **406** may generate the first estimated angle measurement signal θest,x based the first angle measurement signal θmeas,x, in addition to the first angle setpoint θsp,x and the dynamic model X. Using both the first angle setpoint θsp,x and the first angle measurement signal θmeas,x as inputs to the dynamic model X may constitute a type of state observer or Kalman filtering algorithm, where the first estimated angle measurement signal θest,x is an optimal measurement of the state of the first scanner **102***x* about the first scanning axis **112**.

[0061] Thus, the system controller **106** may apply the first angle setpoint θsp,x as a first input to the dynamic model X to obtain a first estimated angle value as a first output from the dynamic model X. The processor **406** may remove angle measurement noise based on the dynamic model X to generate the first estimated angle measurement signal θest,x. Thus, the processor **406** may generate the first estimated angle measurement signal θest,x to be substantially free of angle measurement noise based on the dynamic model X.

[0062] In some implementations, the processor **406** may remove a first angle measurement noise component from the first angle measurement signal θmeas,x to generate the first estimated angle measurement signal θest,x based on applying the first angle setpoint θsp,x and the first angle measurement signal θmeas,x as first inputs to the dynamic model X. For example, the processor **406** may, based on receiving the first angle setpoint θsp,x as a first input to the dynamic model X, attenuate a first angle measurement noise from the first angle measurement signal θmeas,x to generate the first estimated angle measurement signal θest,x. In some implementations, the processor **406** may include a noise attenuating filter programmed by the dynamic model X, and the noise attenuating filter may output the first estimated angle measurement signal θest,x based on the first angle setpoint θsp,x.

[0063] The second processing system **404** may include a control loop **200***y* that is similar to the control loop **200** described in connection with FIG. **2**. The control loop **200***y* may be configured for driving a galvanometer scanner about the second scanning axis **114** and obtaining a second estimated angle measurement signal θest,y that may be provided to a processing component of the processing circuitry **118** for generating point cloud data. The galvanometer scanner may be the scanner **102** or the second scanner **102***y*. In the following examples, it will be assumed that the galvanometer scanner controlled by the control loop **200***y* corresponds to the second scanner **102***y*.

[0064] A driver system (e.g., driver system **104**) of the control loop **200***y* may receive a second angle setpoint θsp,y for the second scanner **102***y*, and drive the second scanner **102***y* with a second driving signal based on the second angle setpoint θsp,y. A second angle position detector of the control loop **200***y* may generate a second angle measurement signal θmeas,y based on detecting a second angular position of the second scanner **102***y* about the second scanning axis **114**. The second angle measurement signal θmeas,y may be a noisy measurement signal.

[0065] A processor **408**, implemented in the system controller **106**, may be configured with a dynamic model Y that models the behavior of the second scanner **102***y* about the second scanning axis **114**. The processor **406** and the processor **408** may be a same processor or may be separate processors. Likewise the dynamic model X and the dynamic model Y may be part of a larger system model or may be separate system models that represent a dynamic model of the two-dimensional scanner.

[0066] The processor **408** may generate the second estimated angle measurement signal θest,y based on the second angle setpoint θsp,y and the dynamic model Y. The second estimated angle measurement signal θest,y may follow a second angular trajectory of the second scanner **102***y* about the second scanning axis **114**.

[0067] In some implementations, the processor **408** may generate the second estimated angle measurement signal θest,y based the second angle measurement signal θmeas,y, in addition to the second angle setpoint θsp,y and the dynamic model Y. Using both the second angle setpoint θsp,y and the second angle measurement signal θmeas,y as inputs to the dynamic model Y may constitute a type of state observer or Kalman filtering algorithm, where the second estimated angle measurement signal θest,y is an optimal measurement of the state of the second scanner **102***y* about the second scanning axis **114**.

[0068] Thus, the system controller **106** may apply the second angle setpoint θsp,y as a second input to the dynamic model Y to obtain a second estimated angle value as a second output from the dynamic model Y. The processor **408** may remove angle measurement noise based on the dynamic model Y to generate the second estimated angle measurement signal θest,y. Thus, the processor **408** may generate the second estimated angle measurement signal θest,y to be substantially free of angle measurement noise based on the dynamic model Y.

[0069] In some implementations, the processor **408** may remove a second angle measurement noise component from the second angle measurement signal θmeas,y to generate the second estimated angle measurement signal θest,y based on applying the second angle setpoint θsp,y and the second angle measurement signal θmeas,y as second inputs to the dynamic model Y. For example, the processor **408** may, based on receiving the second angle setpoint θsp,y as a second input to the dynamic model Y, attenuate a second angle measurement noise from the second angle measurement signal θmeas,y to generate the second estimated angle measurement signal θest,y. In some implementations, the processor **408** may include a noise attenuating filter programmed by the dynamic model Y, and the noise attenuating filter may output the second estimated angle measurement signal θest,y based on the second angle setpoint θsp,y.

[0070] The processing circuitry **118** may receive the first estimated angle measurement signal θest,x, the second estimated angle measurement signal θest,y, and a distance measurement signal D. The distance measurement signal D may be provided by the sensor **110** and may include distance measurements obtained from one or more reflected light beams.

[0071] The processing circuitry **118** may associate a distance measurement with a first estimated angle value corresponding to the first estimated angle measurement signal θest,x, associate the distance measurement with a second estimated angle value corresponding to the second estimated angle measurement signal θest,y, and generate a point in a 3D point cloud based on the distance measurement, the first estimated angle value, and the second estimated angle value.

[0072] As indicated above, FIG. **4** is provided as an example. Other examples may differ from what is described with regard to FIG. **4**. In practice, the system **400** may include additional components, fewer components, different components, or differently arranged components than those shown in FIG. **4** without deviating from the disclosure provided above. Additionally, two or more components shown in FIG. **4** may be implemented within a single component, or a single component shown in FIG. **4** may be implemented as multiple, distributed components. Additionally, or alternatively, a set of components (e.g., one or more components) of the system **400** may perform one or more functions described as being performed by another set of components of the system **400**.

[0073] FIG. **5** shows a system **500** according to one or more implementations. The system **500** may include a processing system **502** corresponding to the first scanning axis **112** and the second scanning axis **114**. Thus, the processing system **502** may be used in the 2D scanning system **100**B described in conjunction with FIG. **1**B.

[0074] The processing system **502** may include a control loop **200***xy* that is similar to the control loop **200** described in connection with FIG. **2**, but may operate based on angle vectors. The control loop **200***xy* may be configured for driving a galvanometer scanner about the first scanning axis **112** and about the second scanning axis **114**, and obtaining an estimated angle vector measurement signal θest,xy that may be provided to a processing component of the processing circuitry **118** for generating point cloud data. The galvanometer scanner may be the scanner **102**.

[0075] A driver system (e.g., driver system **104**) of the control loop **200***xy* may receive an angle vector setpoint θvsp corresponding to a two-dimensional scanning coordinate, drive the scanner **102** about the first scanning axis **112** with the first driving signal based on the angle vector setpoint θvsp, and drive the scanner **102** about the second scanning axis **114** with the second driving signal based on the angle vector setpoint θvsp. A first angle position detector of the control loop **200***xy* may generate a first angle measurement signal θmeas,x based on detecting a first angular position of the first scanner **102** about the first scanning axis **112**. A second angle position detector of the control loop **200***xy* may generate a second angle measurement signal θmeas,y based on detecting a second angular position of the scanner **102***y* about the second scanning axis **114**. The first angle measurement signal θmeas,x and the second angle measurement signal θmeas,y may be noisy measurement signals. The control loop **200***xy* may combine the first angle measurement signal θmeas,x and the second angle measurement signal θmeas,y to generate an angle measurement vector signal θmeas,xy, which may also be a noisy signal.

[0076] A processor **504**, implemented in the system controller **106**, may be configured with a dynamic model XY that models the behavior of the scanner **102** about the first scanning axis **112** and the second scanning axis **114**, and may model cross-coupling effects of the first scanning axis **112** and the second scanning axis **114**. The processor **504** may generate a first estimated angle measurement signal θest,x based on the angle vector setpoint θvsp and the dynamic model XY. The first estimated angle measurement signal θest,x may follow a first angular trajectory of the scanner **102** about the first scanning axis **112**. Additionally, the processor **504** may generate a second estimated angle measurement signal θest,y based on the angle vector setpoint θvsp and the dynamic model XY. The second estimated angle measurement signal θest,y may follow a second angular trajectory of the scanner **102** about the second scanning axis **114**.

[0077] In some implementations, the processor **504** may generate the first estimated angle measurement signal θest,x based on the angle vector setpoint θvsp, the first angle measurement signal θmeas,x, and the dynamic model XY. Additionally, the processor **504** may generate the second estimated angle measurement signal θest,y based on the angle vector setpoint θvsp, the second angle measurement signal θmeas,y, and the dynamic mode XY.
[0078] In some implementations, the processor **504** may combine the first estimated angle measurement signal θest,x and the second estimated angle measurement signal θest,y to generate the estimated angle vector measurement signal θest,xy.
[0079] In some implementations, the processor **504** may generate the estimated angle vector measurement signal θest,xy based on the angle vector setpoint θvsp, the angle measurement vector signal θmeas,xy, and the dynamic model XY.
[0080] The system controller **106** may to apply the angle vector setpoint θvsp as an input to the dynamic model XY to obtain a first estimated angle value as a first output from the dynamic model XY and to obtain a second estimated angle value as a second output from the dynamic model XY. The processor **504** may remove angle measurement noise based on the dynamic model XY to generate the first estimated angle measurement signal θest,x and the second estimated angle measurement signal θest,y. The processor **504** may generate the first estimated angle measurement signal θest,x and the second estimated angle measurement signal θest,y substantially free of angle measurement noise based on the dynamic model XY. For example, the processor **504** may remove a first angle measurement noise component from the first angle measurement signal θmeas,x to generate the first estimated angle measurement signal θest,x based on applying the angle vector setpoint θvsp and the first angle measurement signal θmeas,x as first inputs to the dynamic model XY. Additionally, the processor **504** may remove a second angle measurement noise component from the second angle measurement signal θmeas,y to generate the second estimated angle measurement signal θest,y based on applying the angle vector setpoint θvsp and the second angle measurement signal θmeas,y as second inputs to the dynamic model XY. Alternatively, the processor **504** may remove measurement noise from the measurement vector signal θmeas,xy based on the angle vector setpoint θvsp in order to generate the estimated angle vector measurement signal θest,xy.
[0081] In some implementations, the processor **504** may, based on receiving the angle vector setpoint θvsp as an input to the dynamic model XY, attenuate a first angle measurement noise from the first angle measurement signal θmeas,x to generate the first estimated angle measurement signal θest,x, and attenuate a second angle measurement noise from the second angle measurement signal θmeas,y to generate the second estimated angle measurement signal θest,x. For example, the processor **504** may include a noise attenuating filter programmed by the dynamic model XY, and the noise attenuating filter may output the first estimated angle measurement signal θest,x and the second estimated angle measurement signal θest,y based on the angle vector setpoint θvsp.
[0082] In some implementations, the processor **504** may, based on receiving the angle vector setpoint θvsp as an input to the dynamic model XY, attenuate angle measurement noise from the measurement vector signal θmeas,xy to generate the estimated angle vector measurement signal θest,xy. For example, the processor **504** may include a noise attenuating filter programmed by the dynamic model XY based on the angle vector setpoint θvsp.
[0083] The processing circuitry **118** may receive the first estimated angle measurement signal θest,x, the second estimated angle measurement signal θest,y, and a distance measurement signal D. The distance measurement signal D may be provided by the sensor **110** and may include distance measurements obtained from one or more reflected light beams.
[0084] The processing circuitry **118** may associate a distance measurement with a first estimated angle value corresponding to the first estimated angle measurement signal θest,x, associate the distance measurement with a second estimated angle value corresponding to the second estimated angle measurement signal θest,y, and generate a point in a 3D point cloud based on the distance

measurement, the first estimated angle value, and the second estimated angle value.
[0085] Alternatively, the processing circuitry **118** may receive the estimated angle vector measurement signal θest,xy, associate a distance measurement with a first estimated angle value corresponding to the estimated angle vector measurement signal θest,xy, associate the distance measurement with a second estimated angle value corresponding to the estimated angle vector measurement signal θest,xy, and generate a point in a 3D point cloud based on the distance measurement, the first estimated angle value, and the second estimated angle value.
[0086] As indicated above, FIG. **5** is provided as an example. Other examples may differ from what is described with regard to FIG. **5**. In practice, the system **500** may include additional components, fewer components, different components, or differently arranged components than those shown in FIG. **5** without deviating from the disclosure provided above. Additionally, two or more components shown in FIG. **5** may be implemented within a single component, or a single component shown in FIG. **5** may be implemented as multiple, distributed components. Additionally, or alternatively, a set of components (e.g., one or more components) of the system **500** may perform one or more functions described as being performed by another set of components of the system **500**.
[0087] FIG. **6** is a flowchart of an example process **600** associated with a beam scanning method. The beam scanning method may include a method for attenuating effects of galvanometer angle measurement noise in a scanner point cloud. In some implementations, one or more process blocks of FIG. **6** are performed by a beam scanning system (e.g., system **400** or system **500**). For example, one or more process blocks of FIG. **6** may be performed by one or more components of system **400** or system **500**, which may include modeling system **300**.
[0088] The example process **600** may include generating a first driving signal based on a first angle setpoint (block **610**); generating a second driving signal based on a second angle setpoint (block **620**); driving a two-dimensional scanner about a first scanning axis based on the first driving signal and about a second scanning axis based on the second driving signal (block **630**); generating a distance measurement based on a reflected light beam (block **640**); generating a first estimated angle measurement signal based on the first angle setpoint and a dynamic model of the two-dimensional scanner, wherein the first estimated angle measurement signal follows a first angular trajectory about the first scanning axis (block **650**); generating a second estimated angle measurement signal based on the second angle setpoint and the dynamic model of the two-dimensional scanner, wherein the second estimated angle measurement signal follows a second angular trajectory about the second scanning axis (block **660**); associating the distance measurement with a first estimated angle value corresponding to the first estimated angle measurement signal (block **670**); associating the distance measurement with a second estimated angle value corresponding to the second estimated angle measurement signal (block **680**); and generating a point in a three-dimensional point cloud based on the distance measurement, the first estimated angle value, and the second estimated angle value (block **690**).
[0089] Process **600** may include additional implementations, such as any single implementation or any combination of implementations described below and/or in connection with one or more other processes described elsewhere herein.
[0090] Although FIG. **6** shows example blocks of process **600**, in some implementations, process **600** includes additional blocks, fewer blocks, different blocks, or differently arranged blocks than those depicted in FIG. **6**. Additionally, or alternatively, two or more of the blocks of process **600** may be performed in parallel.
[0091] The following provides an overview of some Aspects of the present disclosure:
[0092] Aspect 1: A beam scanning system, comprising: a two-dimensional scanner comprising a first galvanometer scanner configured to rotate about a first scanning axis based on a first driving signal, and a second galvanometer scanner configured to rotate about a second scanning axis based on a second driving signal; a time-of-flight sensor configured to receive a reflected light beam and

generate a distance measurement based on the reflected light beam; a driver system configured to receive a first angle setpoint for the first galvanometer scanner and a second angle setpoint for the second galvanometer scanner, drive the first galvanometer scanner with the first driving signal based on the first angle setpoint, and drive the second galvanometer scanner with the second driving signal based on the second angle setpoint; and a system controller configured with a dynamic model of the two-dimensional scanner, wherein the system controller is configured to generate a first estimated angle measurement signal based on the first angle setpoint and the dynamic model, wherein the first estimated angle measurement signal follows a first angular trajectory of the first galvanometer scanner about the first scanning axis, wherein the system controller is configured to generate a second estimated angle measurement signal based on the second angle setpoint and the dynamic model, wherein the second estimated angle measurement signal follows a second angular trajectory of the second galvanometer scanner about the second scanning axis, and wherein the system controller is configured to associate the distance measurement with a first estimated angle value corresponding to the first estimated angle measurement signal, associate the distance measurement with a second estimated angle value corresponding to the second estimated angle measurement signal, and generate a point in a three-dimensional point cloud based on the distance measurement, the first estimated angle value, and the second estimated angle value.

[0093] Aspect 2: The beam scanning system of Aspect 1, wherein the system controller is configured to apply the first angle setpoint as a first input to the dynamic model to obtain the first estimated angle value as a first output from the dynamic model, and wherein the system controller is configured to apply the second angle setpoint as a second input to the dynamic model to obtain the second estimated angle value as a second output from the dynamic model.

[0094] Aspect 3: The beam scanning system of any of Aspects 1-2, wherein the system controller is configured to remove angle measurement noise based on the dynamic model to generate the first estimated angle measurement signal and the second estimated angle measurement signal.

[0095] Aspect 4: The beam scanning system of any of Aspects 1-3, wherein the system controller is configured to generate the first estimated angle measurement signal and the second estimated angle measurement signal substantially free of angle measurement noise based on the dynamic model.

[0096] Aspect 5: The beam scanning system of any of Aspects 1-4, further comprising: a first angle position detector configured to generate a first angle measurement signal based on detecting a first angular position of the first galvanometer scanner about the first scanning axis; and a second angle position detector configured to generate a second angle measurement signal based on detecting a second angular position of the second galvanometer scanner about the second scanning axis, wherein the system controller is configured to generate the first estimated angle measurement signal based on the first angle setpoint, the first angle measurement signal, and the dynamic model, and wherein the system controller is configured to generate the second estimated angle measurement signal based on the second angle setpoint, the second angle measurement signal, and the dynamic model.

[0097] Aspect 6: The beam scanning system of Aspect 5, wherein the system controller is configured to remove a first angle measurement noise component from the first angle measurement signal to generate the first estimated angle measurement signal based on applying the first angle setpoint and the first angle measurement signal as first inputs to the dynamic model, and wherein the system controller is configured to remove a second angle measurement noise component from the second angle measurement signal to generate the second estimated angle measurement signal based on applying the second angle setpoint and the second angle measurement signal as second inputs to the dynamic model.

[0098] Aspect 7: The beam scanning system of Aspect 5, wherein the system controller is configured to, based on receiving the first angle setpoint as a first input to the dynamic model, attenuate a first angle measurement noise from the first angle measurement signal to generate the first estimated angle measurement signal, and wherein the system controller is configured to, based on receiving the second angle setpoint as a second input to the dynamic model, attenuate a second angle measurement noise from the second angle measurement signal to generate the second estimated angle measurement signal.

[0099] Aspect 8: The beam scanning system of Aspect 5, wherein the system controller includes a noise attenuating filter programmed by the dynamic model, and wherein the noise attenuating filter is configured to output the first estimated angle measurement signal and the second estimated angle measurement signal.

[0100] Aspect 9: The beam scanning system of any of Aspects 1-8, wherein the driver system is configured to generate the first driving signal based on a difference between the first estimated angle measurement signal and a first setpoint control signal, and wherein the driver system is configured to generate the second driving signal based on a difference between the second estimated angle measurement signal and a second setpoint control signal.

[0101] Aspect 10: A beam scanning system, comprising: a two-dimensional scanner comprising a galvanometer scanner configured to rotate about a first scanning axis based on a first driving signal, and rotate about a second scanning axis based on a second driving signal; a time-of-flight sensor configured to receive a reflected light beam and generate a distance measurement based on the reflected light beam; a driver system configured to receive an angle vector setpoint corresponding to a two-dimensional scanning coordinate, drive the galvanometer scanner about the first scanning axis with the first driving signal based on the angle vector setpoint, and drive the galvanometer scanner about the second scanning axis with the second driving signal based on the angle vector setpoint; and a system controller configured with a dynamic model of the two-dimensional scanner, wherein the system controller is configured to generate a first estimated angle measurement signal based on the angle vector setpoint and the dynamic model, wherein the first estimated angle measurement signal follows a first angular trajectory of the galvanometer scanner about the first scanning axis, wherein the system controller is configured to generate a second estimated angle measurement signal based on the angle vector setpoint and the dynamic model, wherein the second estimated angle measurement signal follows a second angular trajectory of the galvanometer scanner about the second scanning axis, and wherein the system controller is configured to associate the distance measurement with a first estimated angle value corresponding to the first estimated angle measurement signal, associate the distance measurement with a second estimated angle value corresponding to the second estimated angle measurement signal, and generate a point in a three-dimensional point cloud based on the distance measurement, the first estimated angle value, and the second estimated angle value.

[0102] Aspect 11: The beam scanning system of Aspect 10, wherein the system controller is configured to apply the angle vector setpoint as an input to the dynamic model to obtain the first estimated angle value as a first output from the dynamic model and to obtain the second estimated angle value as a second output from the dynamic model.

[0103] Aspect 12: The beam scanning system of any of Aspects 10-11, wherein the system controller is configured to remove angle measurement noise based on the dynamic model to generate the first estimated angle measurement signal and the second estimated angle measurement signal.

[0104] Aspect 13: The beam scanning system of any of Aspects 10-12, wherein the system controller is configured to generate the first estimated angle measurement signal and the second estimated angle measurement signal substantially free of angle measurement noise based on the dynamic model.

[0105] Aspect 14: The beam scanning system of any of Aspects 10-13, further comprising: a first angle position detector configured to generate a first angle measurement signal based on detecting a first angular position of the galvanometer scanner about the first scanning axis; and a second angle position detector configured to generate a second angle measurement signal based on detecting a second angular position of the galvanometer scanner about the second scanning axis, wherein the system controller is configured to generate an estimated angle vector measurement signal based on the angle vector setpoint, an angle measurement vector signal, and the dynamic model, wherein the angle measurement vector signal represents a combination of the first angle measurement signal and the second angle measurement signal, and wherein the estimated angle vector measurement signal represents a combination of the first estimated angle measurement signal and the second estimated angle measurement signal.

[0106] Aspect 15: The beam scanning system of Aspect 14, wherein the system controller is configured to remove a first angle measurement noise component from the first angle measurement signal to generate the first estimated angle measurement signal based on applying the angle vector setpoint and the first angle measurement signal as first inputs to the dynamic model, and wherein the system controller is configured to remove a second angle measurement noise component from the second angle measurement signal to generate the second estimated angle measurement signal based on applying the angle vector setpoint and the second angle measurement signal as second inputs to the dynamic model.

[0107] Aspect 16: The beam scanning system of Aspect 14, wherein the system controller is configured to, based on receiving the angle vector setpoint as an input to the dynamic model, attenuate a first angle measurement noise from the first angle measurement signal to generate the first estimated angle measurement signal, and attenuate a second angle measurement noise from the second angle measurement signal to generate the second estimated angle measurement signal.

[0108] Aspect 17: The beam scanning system of Aspect 14, wherein the system controller includes a noise attenuating filter programmed by the dynamic model, and wherein the noise attenuating filter is configured to output the first estimated angle measurement signal and the second estimated angle measurement signal.

[0109] Aspect 18: The beam scanning system of any of Aspects 10-17, wherein the driver system is configured to compensate the first driving signal based on a difference between the first estimated angle measurement signal and a first setpoint control signal, and wherein the driver system is configured to compensate the second driving signal based on a difference between the second estimated angle measurement signal and a second setpoint control signal.

[0110] Aspect 19: The beam scanning system of any of Aspects 10-18, wherein the angle vector setpoint includes a first angle setpoint for the first scanning axis and a second angle setpoint for the second scanning axis.

[0111] Aspect 20: A beam scanning method, comprising: generating a first driving signal based on a first angle setpoint; generating a second driving signal based on a second angle setpoint; driving a two-dimensional scanner about a first scanning axis based on the first driving signal and about a second scanning axis based on the second driving signal; generating a distance measurement based on a reflected light beam; generating a first estimated angle measurement signal based on the first angle setpoint and a dynamic model of the two-dimensional scanner, wherein the first estimated angle measurement signal follows a first angular trajectory about the first scanning axis; generating a second estimated angle measurement signal based on the second angle setpoint and the dynamic model of the two-dimensional scanner, wherein the second estimated angle measurement signal follows a second angular trajectory about the second scanning axis; associating the distance measurement with a first estimated angle value corresponding to the first estimated angle measurement signal; associating the distance measurement with a second estimated angle value corresponding to the second estimated angle measurement signal; and generating a point in a three-dimensional point cloud based on the distance measurement, the first estimated angle value, and the second estimated angle value.

[0112] Aspect 21: A system configured to perform one or more operations recited in one or more of Aspects 1-20.

[0113] Aspect 22: An apparatus comprising means for performing one or more operations recited in one or more of Aspects 1-20.

[0114] Aspect 23: A non-transitory computer-readable medium storing a set of instructions, the set of instructions comprising one or more instructions that, when executed by a device, cause the device to perform one or more operations recited in one or more of Aspects 1-20.

[0115] Aspect 24: A computer program product comprising instructions or code for executing one or more operations recited in one or more of Aspects 1-20.

[0116] The foregoing disclosure provides illustration and description, but is not intended to be exhaustive or to limit the implementations to the precise forms disclosed. Modifications and variations may be made in light of the above disclosure or may be acquired from practice of the implementations. Furthermore, any of the implementations described herein may be combined unless the foregoing disclosure expressly provides a reason that one or more implementations may not be combined.

[0117] As used herein, the term “component” is intended to be broadly construed as hardware, firmware, and/or a combination of hardware and software. It will be apparent that systems and/or methods described herein may be implemented in different forms of hardware, firmware, or a combination of hardware and software. The actual specialized control hardware or software code used to implement these systems and/or methods is not limiting of the implementations. Thus, the operation and behavior of the systems and/or methods are described herein without reference to specific software code—it being understood that software and hardware can be designed to implement the systems and/or methods based on the description herein.

[0118] Even though particular combinations of features are recited in the claims and/or disclosed in the specification, these combinations are not intended to limit the disclosure of various implementations. In fact, many of these features may be combined in ways not specifically recited in the claims and/or disclosed in the specification. Although each dependent claim listed below may directly depend on only one claim, the disclosure of various implementations includes each dependent claim in combination with every other claim in the claim set. As used herein, a phrase referring to “at least one of” a list of items refers to any combination of those items, including single members. As an example, “at least one of: a, b, or c” is intended to cover a, b, c, a-b, a-c, b-c, and a-b-c, as well as any combination with multiple of the same item.

[0119] When a component or one or more components (e.g., a laser emitter or one or more laser emitters) is described or claimed (within a single claim or across multiple claims) as performing multiple operations or being configured to perform multiple operations, this language is intended to broadly cover a variety of architectures and environments. For example, unless explicitly claimed otherwise (e.g., via the use of “first component” and “second component” or other language that differentiates components in the claims), this language is intended to cover a single component performing or being configured to perform all of the operations, a group of components collectively performing or being configured to perform all of the operations, a first component performing or being configured to perform a first operation and a second component performing or being configured to perform a second operation, or any combination of components performing or being configured to perform the operations. For example, when a claim has the form “one or more components configured to: perform X; perform Y; and perform Z,” that claim should be interpreted to mean “one or more components configured to perform X; one or more (possibly different) components configured to perform Y; and one or more (also possibly different) components configured to perform Z.”

[0120] No element, act, or instruction used herein should be construed as critical or essential unless explicitly described as such. Also, as used herein, the articles “a” and “an” are intended to include one or more items, and may be used interchangeably with “one or more.” Further, as used herein, the article “the” is intended to include one or more items referenced in connection with the article “the” and may be used interchangeably with “the one or more.” Furthermore, as used herein, the term “set” is intended to include one or more items (e.g., related items, unrelated items, or a combination of related and unrelated items), and may be used interchangeably with “one or more.” Where only one item is intended, the phrase “only one” or similar language is used. Also, as used herein, the terms “has,” “have,” “having,” or the like are intended to be open-ended terms. Further, the phrase “based on” is intended to mean “based, at least in part, on” unless explicitly stated otherwise. Also, as used herein, the term “or” is intended to be inclusive when used in a series and may be used interchangeably with “and/or,” unless explicitly stated otherwise (e.g., if used in combination with “either” or “only one of”).

Claims

1. A beam scanning system, comprising: a two-dimensional scanner comprising a first galvanometer scanner configured to rotate about a first scanning axis based on a first driving signal, and a second galvanometer scanner configured to rotate about a second scanning axis based on a second driving signal; a time-of-flight sensor configured to receive a reflected light beam and generate a distance measurement based on the reflected light beam; a driver system configured to receive a first angle setpoint for the first galvanometer scanner and a second angle setpoint for the second galvanometer scanner, drive the first galvanometer scanner with the first driving signal based on the first angle setpoint, and drive the second galvanometer scanner with the second driving signal based on the second angle setpoint; and a system controller configured with a dynamic model of the two-dimensional scanner, wherein the system controller is configured to generate a first estimated angle measurement signal based on the first angle setpoint and the dynamic model, wherein the first estimated angle measurement signal follows a first angular trajectory of the first galvanometer scanner about the first scanning axis, wherein the system controller is configured to generate a second estimated angle measurement signal based on the second angle setpoint and the dynamic model, wherein the second estimated angle measurement signal follows a second angular trajectory of the second galvanometer scanner about the second scanning axis, and wherein the system controller is configured to associate the distance measurement with a first estimated angle value corresponding to the first estimated angle measurement signal, associate the distance measurement with a second estimated angle value corresponding to the second estimated angle measurement signal, and generate a point in a three-dimensional point cloud based on the distance measurement, the first estimated angle value, and the second estimated angle value.
2. The beam scanning system of claim 1, wherein the system controller is configured to apply the first angle setpoint as a first input to the dynamic model to obtain the first estimated angle value as a first output from the dynamic model, and wherein the system controller is configured to apply the second angle setpoint as a second input to the dynamic model to obtain the second estimated angle value as a second output from the dynamic model.
3. The beam scanning system of claim 1, wherein the system controller is configured to remove angle measurement noise based on the dynamic model to generate the first estimated angle measurement signal and the second estimated angle measurement signal.
4. The beam scanning system of claim 1, wherein the system controller is configured to generate the first estimated angle measurement signal and the second estimated angle measurement signal substantially free of angle measurement noise based on the dynamic model.
5. The beam scanning system of claim 1, further comprising: a first angle position detector configured to generate a first angle measurement signal based on detecting a first angular position of the first galvanometer scanner about the first scanning axis; and a second angle position detector configured to generate a second angle measurement signal based on detecting a second angular position of the second galvanometer scanner about the second scanning axis, wherein the system controller is configured to generate the first estimated angle measurement signal based on the first angle setpoint, the first angle measurement signal, and the dynamic model, and wherein the system

controller is configured to generate the second estimated angle measurement signal based on the second angle setpoint, the second angle measurement signal, and the dynamic model.

6. The beam scanning system of claim 5, wherein the system controller is configured to remove a first angle measurement noise component from the first angle measurement signal to generate the first estimated angle measurement signal based on applying the first angle setpoint and the first angle measurement signal as first inputs to the dynamic model, and wherein the system controller is configured to remove a second angle measurement noise component from the second angle measurement signal to generate the second estimated angle measurement signal based on applying the second angle setpoint and the second angle measurement signal as second inputs to the dynamic model.

7. The beam scanning system of claim 5, wherein the system controller is configured to, based on receiving the first angle setpoint as a first input to the dynamic model, attenuate a first angle measurement noise from the first angle measurement signal to generate the first estimated angle measurement signal, and wherein the system controller is configured to, based on receiving the second angle setpoint as a second input to the dynamic model, attenuate a second angle measurement noise from the second angle measurement signal to generate the second estimated angle measurement signal.

8. The beam scanning system of claim 5, wherein the system controller includes a noise attenuating filter programmed by the dynamic model, and wherein the noise attenuating filter is configured to output the first estimated angle measurement signal and the second estimated angle measurement signal.

9. The beam scanning system of claim 1, wherein the driver system is configured to generate the first driving signal based on a difference between the first estimated angle measurement signal and a first setpoint control signal, and wherein the driver system is configured to generate the second driving signal based on a difference between the second estimated angle measurement signal and a second setpoint control signal.

10. A beam scanning system, comprising: a two-dimensional scanner comprising a galvanometer scanner configured to rotate about a first scanning axis based on a first driving signal, and rotate about a second scanning axis based on a second driving signal; a time-of-flight sensor configured to receive a reflected light beam and generate a distance measurement based on the reflected light beam; a driver system configured to receive an angle vector setpoint corresponding to a two-dimensional scanning coordinate, drive the galvanometer scanner about the first scanning axis with the first driving signal based on the angle vector setpoint, and drive the galvanometer scanner about the second scanning axis with the second driving signal based on the angle vector setpoint; and a system controller configured with a dynamic model of the two-dimensional scanner, wherein the system controller is configured to generate a first estimated angle measurement signal based on the angle vector setpoint and the dynamic model, wherein the first estimated angle measurement signal follows a first angular trajectory of the galvanometer scanner about the first scanning axis, wherein the system controller is configured to generate a second estimated angle measurement signal based on the angle vector setpoint and the dynamic model, wherein the second estimated angle measurement signal follows a second angular trajectory of the galvanometer scanner about the second scanning axis, and wherein the system controller is configured to associate the distance measurement with a first estimated angle value corresponding to the first estimated angle measurement signal, associate the distance measurement with a second estimated angle value corresponding to the second estimated angle measurement signal, and generate a point in a three-dimensional point cloud based on the distance measurement, the first estimated angle value, and the second estimated angle value.

11. The beam scanning system of claim 10, wherein the system controller is configured to apply the angle vector setpoint as an input to the dynamic model to obtain the first estimated angle value as a first output from the dynamic model and to obtain the second estimated angle value as a second output from the dynamic model.

12. The beam scanning system of claim 10, wherein the system controller is configured to remove angle measurement noise based on the dynamic model to generate the first estimated angle measurement signal and the second estimated angle measurement signal.

13. The beam scanning system of claim 10, wherein the system controller is configured to generate the first estimated angle measurement signal and the second estimated angle measurement signal substantially free of angle measurement noise based on the dynamic model.

14. The beam scanning system of claim 10, further comprising: a first angle position detector configured to generate a first angle measurement signal based on detecting a first angular position of the galvanometer scanner about the first scanning axis; and a second angle position detector configured to generate a second angle measurement signal based on detecting a second angular position of the galvanometer scanner about the second scanning axis, wherein the system controller is configured to generate an estimated angle vector measurement signal based on the angle vector setpoint, an angle measurement vector signal, and the dynamic model, wherein the angle measurement vector signal represents a combination of the first angle measurement signal and the second angle measurement signal, and wherein the estimated angle vector measurement signal represents a combination of the first estimated angle measurement signal and the second estimated angle measurement signal.

15. The beam scanning system of claim 14, wherein the system controller is configured to remove a first angle measurement noise component from the first angle measurement signal to generate the first estimated angle measurement signal based on applying the angle vector setpoint and the first angle measurement signal as first inputs to the dynamic model, and wherein the system controller is configured to remove a second angle measurement noise component from the second angle measurement signal to generate the second estimated angle measurement signal based on applying the angle vector setpoint and the second angle measurement signal as second inputs to the dynamic model.

16. The beam scanning system of claim 14, wherein the system controller is configured to, based on receiving the angle vector setpoint as an input to the dynamic model, attenuate a first angle measurement noise from the first angle measurement signal to generate the first estimated angle measurement signal, and attenuate a second angle measurement noise from the second angle measurement signal to generate the second estimated angle measurement signal.

17. The beam scanning system of claim 14, wherein the system controller includes a noise attenuating filter programmed by the dynamic model, and wherein the noise attenuating filter is configured to output the first estimated angle measurement signal and the second estimated angle measurement signal.

18. The beam scanning system of claim 10, wherein the driver system is configured to compensate the first driving signal based on a difference between the first estimated angle measurement signal and a first setpoint control signal, and wherein the driver system is configured to compensate the second driving signal based on a difference between the second estimated angle measurement signal and a second setpoint control signal.

19. The beam scanning system of claim 10, wherein the angle vector setpoint includes a first angle setpoint for the first scanning axis and a second angle setpoint for the second scanning axis.

20. A beam scanning method, comprising: generating a first driving signal based on a first angle setpoint; generating a second driving signal based on a second angle setpoint; driving a two-dimensional scanner about a first scanning axis based on the first driving signal and about a second scanning axis based on the second driving signal; generating a distance measurement based on a reflected light beam; generating a first estimated angle measurement signal based on the first angle setpoint and a dynamic model of the two-dimensional scanner, wherein the first estimated angle measurement signal follows a first angular trajectory about the first scanning axis; generating a second estimated angle measurement signal based on the second angle setpoint and the dynamic model of the two-dimensional scanner, wherein the second estimated angle measurement signal follows a second angular trajectory about the second scanning axis; associating the distance measurement with a first estimated angle value corresponding to the first estimated angle measurement signal; associating the distance measurement with a second estimated angle value corresponding to the second estimated angle measurement signal; and generating a point in a three-dimensional point cloud based on the distance measurement, the first estimated angle value, and the second estimated angle value.